

SMARTBRIDGE
SKILL WALL

No. of Functional Features Included in the Solution

Date: 30 September 2026
Team ID: SWTID-2026-3226
Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks: 5 Marks

Functional Features Overview:

This document lists the functional features included in the FitBuddy project solution. Each feature describes its purpose, implementation module/component, status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Profile Input	Collects name, user ID, age, weight, goal, intensity and experience level through a validated form.	templates/index.html / schemas.py	Done	0.75
2	Workout Plan Generation	Creates a safe, personalized 7-day workout plan from the user's profile using Gemini-generated output.	gemini_generator.py / routes.py	Done	0.75
3	Nutrition & Recovery Tip	Generates a short goal-based nutrition and recovery tip using a Gemini Flash model.	gemini_flash_generator.py	Done	0.50
4	Feedback-Based Plan Update	Revises the existing 7-day plan according to the user's feedback while keeping safe progression.	updated_plan.py / routes.py	Done	0.75
5	Plan Result Display	Shows the workout plan, nutrition tip, Gemini or Demo mode label and update messages in a web page.	templates/result.html + static/css	Done	0.75
6	User & Plan Storage	Stores user profiles, original and updated plans, feedback and nutrition tips in a SQLite database.	database.py / models.py	Done	0.50
7	Gemini API Integration	Connects the application to Google Gemini through a centralized client, environment-based API key and model settings, with demo-mode fallback.	gemini_client.py / gemini_generator.py / gemini_flash_generator.py	Done	0.50
8	Admin View & REST API	Provides an admin page listing all users and plans, plus JSON endpoints for generating plans, submitting feedback and listing users.	routes.py / templates/all_users.html	Done	0.50

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features

User login, mobile-responsive styling, detailed meal plans, progress tracking, PDF export of plans, and future multilingual/customization support.

Features Tested & Verified

Automated pytest tests cover home page loading, workout plan generation, feedback-based plan update and user listing through the API in demo mode. Gemini-generated plans are stored in the project database, and error handling covers input validation, missing users or plans, and Gemini failures through demo-mode fallback.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Fitness profile form, plan result display and feedback form.
2	Backend / Logic	3	Workout plan generation, nutrition tip generation and feedback-based plan updating.
3	Database / Storage	1	SQLite storage of users, plans and feedback through SQLAlchemy models.
4	API / Integration	2	Google Gemini API through a centralized client, and JSON API routes with an admin view.
5	Security / Configuration	1	GEMINI_API_KEY is loaded through environment configuration and user input is validated with Pydantic; secrets are not hard-coded in application source.

FitBuddy Solution Focus

FitBuddy is an AI-powered fitness plan generator using Gemini models. The solution accepts a user's fitness profile, generates a personalized 7-day workout plan and a nutrition and recovery tip, lets users submit feedback to revise their plan, stores users and plans in a SQLite database, and falls back to demo mode when Gemini is unavailable.

Project Identification

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